FIN2020 Homework 9

Problem 1

Consider a two-date economy where there are three states of the world at date 1. Consumption per capita at date 1 will be \$5, \$10, or \$15. A security that pays one unit of consumption next period is worth \$8 today. The risk-free security pays a gross return of 1.1. A call option on per capita consumption with an exercise price of \$12 costs \$1. The probabilities of the three states are 0.3, 0.4, and 0.3, respectively.

a. Are markets complete? If yes, what are the state prices?

b. Price a put option on per capita consumption with an exercise price of \$8.

c. Can you derive the risk-neutral probabilities?

Problem 2

Suppose that a one-year discount bond that pays off one dollar for sure one year from now sells for $P_1 = 0.90$ today, a two-year discount bond that pays off one dollar for sure two years from now sells for $P_2 = 0.80$ today, and a three-year discount bond that pays off one dollar for sure three years from now sells for $P_3 = 0.70$ today.

a. Consider a three-year coupon bond that makes an annual interest (coupon) payment of 100 dollars at the end of each of the next three years and also returns an additional amount (face value) of 1000 dollars at the end of the third year. What will the price of this bond be if there are no arbitrage opportunities across the markets for discount and coupon bonds?

b. Next, consider a risk-free asset that pays the holder 100 dollars for sure two years from now and 100 dollars for sure three years from now. What will the price of this asset be if there are no arbitrage opportunities across markets for all risk-free assets?

c. Finally, consider another risk-free asset that pays the holder 100 dollars for sure two years from now but then requires the holder *to pay* 100 dollars for sure three years from now. What will the price of this asset be if there are no arbitrage opportunities across markets for all risk-free assets?

Problem 3

Begin with the fundamental equation $E[mR_i] = 1$ where R_i is the gross return on asset i, and m is the pricing kernel. Noting that this fundamental equation should hold for the gross returns on the riskless asset and on the market portfolio (R_M) as well, obtain a CAPM-like expression. Define a functional relation between m and R_M so that you will obtain precisely the CAPM expression.